Quadratic stabilization of the canonical Foulkes–Howe map

Rewritten for readability from the OpenAI manuscript of the same title

Abstract

For every finite-dimensional complex vector space V, the canonical Foulkes–Howe map $\mathrm{Sym}^b(\mathrm{Sym}^a V) \longrightarrow \mathrm{Sym}^a(\mathrm{Sym}^b V)$ is surjective whenever $a \geq 2$ and $b \geq a(a-1)$. This is a quadratic stabilization bound independent of $\dim V$.

1 The result and how it is proved

1.1 Setting

Throughout, V is a finite-dimensional complex vector space and $\mathrm{Sym}\, V = \bigoplus_{k\geq 0} \mathrm{Sym}^k V$ is its symmetric algebra. A product of vectors always means their product in $\mathrm{Sym}\, V$. An element $u_1 \cdots u_k \in \mathrm{Sym}^k V$ with $u_i \in V$ is called a *product of k vectors* (the vectors may repeat or be zero), and u^k is a *pure power*.

Averaging. Let W be a complex vector space and $a \geq 1$ an integer. The symmetric group $\mathfrak{S}_a$ acts on $W^{\otimes a}$ by permuting tensor factors. Define the linear map $\mathrm{Av}\colon W^{\otimes a} \to W^{\otimes a}$ by

$$\mathrm{Av}(w_1 \otimes \cdots \otimes w_a) = \frac{1}{a!} \sum_{\pi \in \mathfrak{S}_a} w_{\pi(1)} \otimes \cdots \otimes w_{\pi(a)}. \tag{1}$$

We use three facts about it.

(i) $\mathrm{Av}\,\tau$ is $\mathfrak{S}_a$-invariant for every τ, and $\mathrm{Av}\,\tau = \tau$ if τ is invariant (all $a!$ permuted copies of τ equal τ). Hence Av is a projection onto $(W^{\otimes a})^{\mathfrak{S}_a}$, and the averages of simple tensors span $(W^{\otimes a})^{\mathfrak{S}_a}$.

(ii) The right side of (1) is symmetric in $(w_1, \ldots, w_a)$, so Av factors through the quotient map $\rho\colon W^{\otimes a} \to \mathrm{Sym}^a W$: there is a linear map $\overline{\mathrm{Av}}\colon \mathrm{Sym}^a W \to (W^{\otimes a})^{\mathfrak{S}_a}$ with $\overline{\mathrm{Av}}(w_1 \cdots w_a) = \mathrm{Av}(w_1 \otimes \cdots \otimes w_a)$.

(iii) $\overline{\mathrm{Av}}$ is an isomorphism with inverse $\rho|_{(W^{\otimes a})^{\mathfrak{S}_a}}$. Indeed, $\rho(\mathrm{Av}(w_1 \otimes \cdots \otimes w_a)) = \frac{1}{a!}\sum_\pi w_{\pi(1)} \cdots w_{\pi(a)} = w_1 \cdots w_a$, and for invariant τ, $\overline{\mathrm{Av}}(\rho(\tau)) = \mathrm{Av}(\tau) = \tau$ by (i).

We call $\overline{\mathrm{Av}}$ the *averaging isomorphism*. Note that $\overline{\mathrm{Av}}(w^a) = w^{\otimes a}$.

The algebra R and the map. Fix an integer $a \geq 1$. Let $\mathcal{P} = (\mathrm{Sym}\, V)^{\otimes a}$ with componentwise multiplication $(f_1 \otimes \cdots \otimes f_a)(g_1 \otimes \cdots \otimes g_a) = f_1 g_1 \otimes \cdots \otimes f_a g_a$. It is a commutative algebra on which $\mathfrak{S}_a$ acts by algebra automorphisms. (The source calls this algebra P_a; it is renamed here so that P can denote a product of vectors.) For $j \geq 0$ put

$$R_j = \big((\mathrm{Sym}^j V)^{\otimes a}\big)^{\mathfrak{S}_a} \subseteq \mathcal{P}, \qquad R = \bigoplus_{j \geq 0} R_j.$$

A product of invariants is invariant and $\mathrm{Sym}^i V \cdot \mathrm{Sym}^j V \subseteq \mathrm{Sym}^{i+j} V$, so $R_i R_j \subseteq R_{i+j}$. Thus R is a graded commutative algebra. Let $A \subseteq R$ be the subalgebra generated by R_1 and put $A_b = A \cap R_b$.

Since a product of j elements of R_1 lies in R_j, the space A_b is the span of all products of exactly b elements of R_1.

The averaging isomorphism with $W = V$ and with $W = \operatorname{Sym}^b V$ gives $R_1 \cong \operatorname{Sym}^a V$ and $R_b \cong \operatorname{Sym}^a(\operatorname{Sym}^b V)$. For $b \geq 0$, the multiplication $(z_1, \ldots, z_b) \mapsto z_1 \cdots z_b$ is a symmetric multilinear map $R_1^b \to R_b$, so it induces a linear map $\operatorname{Sym}^b R_1 \to R_b$ with image A_b. Transported through the two averaging isomorphisms, this is the *canonical Foulkes–Howe map*

$$\mu_{a,b,V} : \operatorname{Sym}^b(\operatorname{Sym}^a V) \longrightarrow \operatorname{Sym}^a(\operatorname{Sym}^b V), \qquad \operatorname{im} \mu_{a,b,V} = A_b. \tag{2}$$

Concretely, u^a corresponds to $u^{\otimes a} \in R_1$ and $(u_1^{\otimes a}) \cdots (u_b^{\otimes a}) = (u_1 \cdots u_b)^{\otimes a}$, so $\mu_{a,b,V}\big((u_1^a) \cdots (u_b^a)\big) = (u_1 \cdots u_b)^a$.

Theorem 1. *For every integer $a \geq 2$, every integer $b \geq a(a-1)$, and every finite-dimensional complex vector space V, the map $\mu_{a,b,V}$ in (2) is surjective. Equivalently, $A_b = R_b$.*

For $a = 1$ we have $R = \operatorname{Sym} V$, and $\mu_{1,b,V}$ is the identity map in every degree.

1.2 Proof at a glance

Key idea. By duality (Proposition 6), surjectivity of $\mu_{a,b,V}$ is equivalent to the following statement: *a symmetric a-linear form T on $\operatorname{Sym}^b V$ with $T(P, \ldots, P) = 0$ for every product P of b vectors is zero.* This is proved by induction on a (Proposition 8). The tool is Lemma 7: the directional-derivative operator $D_{Y,U} = \sum_k Y_k \, \partial/\partial U_k$ is injective on polynomials whose degree in the block U exceeds their degree in the block Y. The lemma is used backwards. To show that a polynomial is zero, we apply a chain of such operators until we reach a polynomial already known to be zero, and we check the degree inequality before each operator.

The inductive step. Put $r = a - 1$ and $m = b - r$; the hypothesis $b \geq a(a-1)$ says $m \geq r^2$. Let Q be a product of m vectors and $x, t \in V$.

Step 1. Over $\mathbb{C}$, $x^r + ct^r$ is a product of r vectors, so $(x^r + ct^r)Q$ is a product of b vectors for every $c \in \mathbb{C}$. The coefficient of c in the diagonal value gives $T(x^rQ, \ldots, x^rQ, t^rQ) = 0$.

Step 2. A chain of m operators, each with a different target block y_i, turns $T(x^rQ, \ldots, x^rQ, t^b)$ into a multiple of the expression in Step 1. Lemma 7 gives $T(x^rQ, \ldots, x^rQ, t^b) = 0$.

Step 3. Step 2 is the diagonal hypothesis for the symmetric r-linear form $(Q_1, \ldots, Q_r) \mapsto T(x^rQ_1, \ldots, x^rQ_r, t^b)$ on $\operatorname{Sym}^m V$. The induction hypothesis (which needs $m \geq r(r-1)$ and $m \geq 1$) shows that this form is zero.

Step 4. A chain of r^2 operators, all with the same target block x, turns $T(v_1^b, \ldots, v_r^b, t^b)$ into a multiple of $T(x^r v_1^m, \ldots, x^r v_r^m, t^b)$, which is zero by Step 3. Before each operator the x-degree is at most $r^2 - 1$ and the degree of the differentiated block is at least $m + 1 \geq r^2 + 1$, so Lemma 7 gives $T(v_1^b, \ldots, v_r^b, t^b) = 0$. Pure powers span $\operatorname{Sym}^b V$, hence $T = 0$.

In Step 4 the degree of the single shared target x accumulates to r^2; this is where the quadratic bound enters.

Smallest case $a = 2$. Here $r = 1$, $b \geq 2$ and $m = b - 1$. Step 1 gives $T(xQ, tQ) = 0$. Step 2 gives $T(xQ, t^b) = 0$ for products Q of $b - 1$ vectors. Step 3 is the case $a = 1$ (products span $\operatorname{Sym}^{b-1} V$) and gives $T(xQ', t^b) = 0$ for all $Q' \in \operatorname{Sym}^{b-1} V$. Step 4 applies the single operator D_{x,v_1} to $T(v_1^b, t^b)$, producing $b\,T(xv_1^{b-1}, t^b) = 0$ with degrees $b > 0$.

Dependencies.

- Lemma 2 (pure powers span) ⇒ Proposition 6 and the base case and last line of Proposition 8.
- Lemma 3, Convention 4 and Lemma 5 (bookkeeping for polynomials built from T) ⇒ Steps 2 and 4.
- Lemma 7 (differential injectivity) ⇒ Steps 2 and 4.
- Proposition 6 + Proposition 8 ⇒ Theorem 1 ⇒ Corollary 9.

Section 7 collects the source's remarks on the literature; nothing there is used in the proofs.

2 Preliminaries

Lemma 2 (Pure powers span)**.** *For every integer $k \geq 1$, the space $\operatorname{Sym}^k V$ is spanned by products of k vectors, and also by pure powers u^k, $u \in V$.*

Used in: Proposition 6 (with $k = a$), and the base case and final line of Proposition 8 (with $k = b$).

Proof. Products span because $\operatorname{Sym}^k V$ is the image of $V^{\otimes k}$, and simple tensors map to products. For pure powers we prove the polarization identity

$$k!\, u_1 \cdots u_k = \sum_{J \subseteq \{1,\dots,k\}} (-1)^{k-|J|} \Big(\sum_{j \in J} u_j \Big)^k \qquad (u_1, \dots, u_k \in V). \tag{3}$$

First take formal commuting variables $s_1, \dots, s_k$ in place of the u_j. By the multinomial theorem, $(\sum_{j \in J} s_j)^k$ is the sum of $\binom{k}{\beta} s^\beta$ over exponent vectors β with $|\beta| = k$ and support in J. Let β have support S. Its coefficient on the right of (3) is

$$\binom{k}{\beta} \sum_{J \supseteq S} (-1)^{k-|J|} = \binom{k}{\beta} (-1)^{k-|S|} \sum_{J' \subseteq \{1,\dots,k\} \setminus S} (-1)^{|J'|} = \binom{k}{\beta} (-1)^{k-|S|} (1-1)^{k-|S|}.$$

This is zero unless $S = \{1, \dots, k\}$. In that case $\beta = (1, \dots, 1)$ and the coefficient is $k!$. So (3) holds in $\mathbb{C}[s_1, \dots, s_k]$. Applying the algebra homomorphism $\mathbb{C}[s_1, \dots, s_k] \to \operatorname{Sym} V$, $s_j \mapsto u_j$, gives it in $\operatorname{Sym} V$. Since $k! \neq 0$ in $\mathbb{C}$, every product of k vectors is a linear combination of pure powers. □

Lemma 3 (Two facts about complex polynomials)**.** *Let $F \in \mathbb{C}[Z_1, \dots, Z_N]$.*

(a) If F vanishes at every point of $\mathbb{C}^N$, then $F = 0$.

(b) Let U be a subset of the variables and $d \geq 0$. If $F(\lambda U, \mathit{rest}) = \lambda^d F(U, \mathit{rest})$ at every point and for every $\lambda \in \mathbb{C}$, then F is homogeneous of degree d in U.

Proof. (a) Induct on N. Write $F = \sum_j F_j(Z_1, \dots, Z_{N-1}) Z_N^j$. For fixed values of $Z_1, \dots, Z_{N-1}$ this is a univariate polynomial with infinitely many roots, so every F_j vanishes at that point. By induction $F_j = 0$.

(b) Write $F = \sum_e F_e$ with F_e homogeneous of degree e in U. At each point, $\sum_e (\lambda^e - \lambda^d) F_e = 0$ for all λ, so F_e vanishes at that point for $e \neq d$. By (a), $F_e = 0$ for $e \neq d$. □

Convention 4 (Vector variables and polynomials built from T)**.** Assume $n = \dim V > 0$ and fix a basis $e_1, \dots, e_n$ of V. A *vector variable* u is a block of n independent scalar variables $(u_1, \dots, u_n)$, standing for the vector $\sum_k u_k e_k$. We use the same letter for the block and for the vector. Let T be an a-linear form on $\operatorname{Sym}^b V$, and consider an expression

$$F = T(M_1, \dots, M_a),$$

in which each M_s is a product of b vectors, each factor being a fixed vector or one of finitely many vector variables. Expanding every vector variable in the basis and using multilinearity of the product and of T shows that F is a polynomial function of the scalar coordinates. By Lemma 3(a) it is a uniquely determined polynomial, and "$F = 0$ for all vector values" is the same as "$F = 0$ as a polynomial". If a vector variable u occurs d times in total among $M_1, \dots, M_a$ (counted with multiplicity), then replacing u by λu multiplies F by λ^d, so by Lemma 3(b) F is homogeneous of degree d in the block u.

For two vector variables U, Y, define the operator

$$D_{Y,U} = \sum_{k=1}^{n} Y_k \frac{\partial}{\partial U_k}.$$

It lowers the U-degree by one and raises the Y-degree by one.

Lemma 5 (Differentiation rule)**.** *In the situation of Convention 4, let u and y be distinct vector variables. Suppose u occurs in exactly one slot s, where $M_s = u^j z$ with $j \geq 1$ and z a product of $b - j$ vectors not involving u. Then*

$$D_{y,u}\, T(M_1, \dots, M_a) = j\, T(M_1, \dots, M_{s-1},\, u^{j-1} y\, z,\, M_{s+1}, \dots, M_a).$$

Used in: Steps 2 and 4 of Proposition 8.

Proof. Fix values of all vector variables and let $\phi(\epsilon)$ be the value of $F = T(M_1, \dots, M_a)$ after replacing u by $u + \epsilon y$ (and nothing else). By the chain rule, $\phi'(0) = \sum_k y_k\, \partial F/\partial u_k = (D_{y,u}F)$ at the chosen point. On the other hand, the binomial theorem in the commutative algebra $\operatorname{Sym} V$ gives $(u + \epsilon y)^j z = u^j z + j\epsilon\, u^{j-1} yz + \epsilon^2 w(\epsilon)$ with $w(\epsilon) \in \operatorname{Sym}^b V$ polynomial in ϵ. Only slot s changes and T is linear in it, so $\phi(\epsilon) = F + j\epsilon\, T(\dots, u^{j-1} yz, \dots) + \epsilon^2(\cdots)$. Hence $\phi'(0) = j\, T(\dots, u^{j-1} yz, \dots)$. The two sides of the claim agree at every point, hence as polynomials by Lemma 3(a). □

3 The canonical image and diagonal vanishing

This section turns surjectivity of $\mu_{a,b,V}$ into a vanishing statement about symmetric multilinear forms. It is used once, in the proof of Theorem 1.

Proposition 6 (Diagonal criterion)**.** *Let $a, b \geq 1$ be integers. The map $\mu_{a,b,V}$ is surjective if and only if every symmetric a-linear form T on $\operatorname{Sym}^b V$ satisfying $T(P, \dots, P) = 0$ for all products P of b vectors is zero.*

Proof. Put $E = \operatorname{Sym}^b V$, so $R_b = (E^{\otimes a})^{\mathfrak{S}_a}$.

Step A: the elements $z(u) := u^{\otimes a}$, $u \in V$, span R_1. The averaging isomorphism $\operatorname{Sym}^a V \to R_1$ sends u^a to $u^{\otimes a}$, and the pure powers u^a span $\operatorname{Sym}^a V$ by Lemma 2 with $k = a$.

Step B: $A_b = \operatorname{span}_{\mathbb{C}}\{P^{\otimes a} : P$ a product of b vectors$\}$. By definition A_b is spanned by the products $z_1 \cdots z_b$ with $z_i \in R_1$. The product is multilinear, so by Step A it is enough to take $z_i = z(u_i)$. Componentwise multiplication gives $z(u_1) \cdots z(u_b) = (u_1 \cdots u_b)^{\otimes a}$.

Step C: functionals on R_b correspond bijectively to symmetric a-linear forms on E. For $\lambda \in R_b^*$ define

$$T_\lambda(w_1, \dots, w_a) = \lambda\big(\operatorname{Av}(w_1 \otimes \cdots \otimes w_a)\big) \qquad (w_i \in E).$$

This is a symmetric a-linear form. The map $\lambda \mapsto T_\lambda$ is injective because averages of simple tensors span R_b (fact (i) on averaging). It is surjective: given a symmetric a-linear form T, let $\widetilde{T}\colon E^{\otimes a} \to \mathbb{C}$

be its linearization and $\lambda = \widetilde{T}|_{R_b}$. Then $T_\lambda(w_1, \dots, w_a) = \frac{1}{a!} \sum_\pi T(w_{\pi(1)}, \dots, w_{\pi(a)}) = T(w_1, \dots, w_a)$ by symmetry of T.

Step D: $T_\lambda(P, \dots, P) = \lambda(P^{\otimes a})$ for every $P \in E$. Indeed $\mathrm{Av}(P^{\otimes a}) = P^{\otimes a}$.

Step E: conclusion. By Steps B and D, λ annihilates A_b if and only if $T_\lambda(P, \dots, P) = 0$ for every product P of b vectors. By Step C, the annihilator of A_b in R_b^* is zero if and only if every symmetric form with vanishing diagonal on products is zero. A subspace A_b of the finite-dimensional space R_b equals R_b if and only if its annihilator is zero, and $A_b = \operatorname{im} \mu_{a,b,V}$ by (2). □

4 Differential injectivity

The following lemma is the only analytic input. The source notes that it is the positive-weight injectivity in the usual $\mathfrak{sl}_2$ weight strings; compare [6, Section 11.1, Claim 11.4 and equation (11.5)]. The proof below is self-contained apart from the definition of the Fischer inner product. In the source the adjoint operator is called E; it is renamed D' here, because E denotes $\mathrm{Sym}^b V$ in Section 3.

Lemma 7 (Differential injectivity)**.** *Let $n \geq 1$, let $U = (U_1, \dots, U_n)$ and $Y = (Y_1, \dots, Y_n)$ be two disjoint blocks of variables, and let p, q be nonnegative integers. On polynomials homogeneous of degree p in U and degree q in Y, the operator $D_{Y,U}$ is injective if $p > q$. The polynomials may depend on any finite collection of additional variables independent of U and Y.*

Used in: Steps 2 and 4 of Proposition 8, each time on a polynomial of the stated bidegree, with all other blocks as additional variables.

Proof. Plan. We put an inner product on the polynomial ring for which the adjoint of $D := D_{Y,U}$ is $D' := D_{U,Y}$. We compute the commutator $D'D - DD'$, and read off a norm identity that forces $h = 0$ when $Dh = 0$ and $p > q$.

Step 1: the inner product. List all scalar variables present (those of U, of Y, and the additional ones) as $Z_1, \dots, Z_N$. Give $\mathbb{C}[Z_1, \dots, Z_N]$ the Fischer, or apolar, Hermitian inner product [1, Section 2, equation (8) and Proposition 3]:

$$\Big\langle \sum_\alpha a_\alpha Z^\alpha, \sum_\beta b_\beta Z^\beta \Big\rangle = \sum_\alpha \alpha!\, a_\alpha \overline{b_\alpha}, \qquad \alpha! := \prod_k \alpha_k!.$$

Distinct monomials are orthogonal, $\|Z^\alpha\|^2 = \alpha! > 0$, and the form is positive definite.

Step 2: differentiation is adjoint to multiplication. For every i and all polynomials g, h, $\langle \partial_{Z_i} g, h \rangle = \langle g, Z_i h \rangle$. Both sides are linear in g and conjugate-linear in h, so it suffices to take $g = Z^\alpha$, $h = Z^\beta$. Writing e_i for the i-th unit exponent vector, both sides equal $\alpha!$ if $\alpha = \beta + e_i$ and 0 otherwise (on the left, $\alpha_i\,(\alpha - e_i)! = \alpha!$). Taking complex conjugates gives also $\langle Z_i g, h \rangle = \langle g, \partial_{Z_i} h \rangle$.

Step 3: $\langle Dg, h \rangle = \langle g, D'h \rangle$. By Step 2, used twice,

$$\langle Dg, h \rangle = \sum_k \langle Y_k\, \partial_{U_k} g, h \rangle = \sum_k \langle \partial_{U_k} g, \partial_{Y_k} h \rangle = \sum_k \langle g, U_k\, \partial_{Y_k} h \rangle = \langle g, D'h \rangle.$$

Step 4: the commutator. By the product rule,

$$D'D = \sum_{j,k} U_j \partial_{Y_j} Y_k \partial_{U_k} = \sum_k U_k \partial_{U_k} + \sum_{j,k} U_j Y_k \partial_{Y_j} \partial_{U_k},$$
$$DD' = \sum_{j,k} Y_k \partial_{U_k} U_j \partial_{Y_j} = \sum_k Y_k \partial_{Y_k} + \sum_{j,k} Y_k U_j \partial_{U_k} \partial_{Y_j}.$$

Subtracting,

$$D'D - DD' = N_U - N_Y, \qquad N_U = \sum_k U_k \partial_{U_k}, \quad N_Y = \sum_k Y_k \partial_{Y_k}.$$

Step 5: the norm identity. Let h be homogeneous of degree p in U and q in Y. On each of its monomials N_U acts as multiplication by p and N_Y by q, so $(N_U - N_Y)h = (p-q)h$. By Step 3, $\|Dh\|^2 = \langle D'Dh, h\rangle$ (the number is real) and $\|D'h\|^2 = \langle DD'h, h\rangle$. Hence, by Step 4,

$$\|Dh\|^2 - \|D'h\|^2 = \langle (N_U - N_Y)h, h\rangle = (p-q)\|h\|^2. \tag{4}$$

Step 6: conclusion. If $Dh = 0$ and $p > q$, the left side of (4) is $-\|D'h\|^2 \le 0$ and the right side is ≥ 0. So $(p-q)\|h\|^2 = 0$, and $h = 0$ by positive definiteness. □

5 Vanishing on products

Proposition 8. *Let V be a finite-dimensional complex vector space, and let $a \ge 1$ and $b \ge 1$ be integers satisfying $b \ge a(a-1)$. If a symmetric a-linear form T on $\operatorname{Sym}^b V$ satisfies*

$$T(P, \dots, P) = 0 \qquad \text{for every product } P \text{ of } b \text{ vectors}, \tag{5}$$

then $T = 0$.

Proof. Plan. We induct on a, proving the statement for all admissible b and all V at once. In the inductive step, Step 1 extracts one identity from (5), Steps 2 and 4 each use Lemma 7 along a chain of derivatives, and Step 3 applies the induction hypothesis in between.

If $V = 0$, then $\operatorname{Sym}^b V = 0$ because $b \ge 1$, and $T = 0$. So assume $\dim V > 0$ and fix a basis, so that Convention 4 applies.

Base case $a = 1$. T is a linear form on $\operatorname{Sym}^b V$ vanishing on all products of b vectors. These span $\operatorname{Sym}^b V$ (Lemma 2), so $T = 0$.

Inductive step. Let $a \ge 2$, assume the proposition for $a-1$ (all b, all V), and put

$$r = a - 1 \ge 1, \qquad m = b - r. \tag{6}$$

From $b \ge a(a-1) = r(r+1)$ we get

$$m = b - r \ge r^2 \ge 1. \tag{7}$$

Step 1: a split binary form. *Claim:* for all $x, t \in V$ and every product $Q = y_1 \cdots y_m$ of m vectors,

$$T(\underbrace{x^r Q, \dots, x^r Q}_{r \text{ slots}}, t^r Q) = 0. \tag{8}$$

(1a) *For every $c \in \mathbb{C}$, $P_c := (x^r + ct^r)Q$ is a product of b vectors.* In $\mathbb{C}[\xi, \eta]$ (the source writes X, Y for these two scalar variables), if $c \ne 0$ choose $\alpha \in \mathbb{C}$ with $\alpha^r = -c$ and put $\zeta = e^{2\pi i/r}$. Since $\prod_{j=0}^{r-1}(\xi - \zeta^j \omega) = \xi^r - \omega^r$, substituting $\omega = \alpha\eta$ gives

$$\xi^r + c\eta^r = \prod_{j=0}^{r-1} (\xi - \alpha\zeta^j \eta).$$

For $c = 0$ the factorization is ξ^r. Applying the algebra homomorphism $\mathbb{C}[\xi, \eta] \to \operatorname{Sym} V$, $\xi \mapsto x$, $\eta \mapsto t$ (no independence of x and t is needed) shows that $x^r + ct^r$ is a product of r vectors of V. Multiplying by Q gives a product of $r + m = b$ vectors.

(1b) *Extracting the coefficient of c.* By (1a) and (5), $T(P_c, \dots, P_c) = 0$ for all $c \in \mathbb{C}$. Write $P_c = x^r Q + c\, t^r Q$. By multilinearity and symmetry,

$$T(P_c, \dots, P_c) = \sum_{k=0}^{a} \binom{a}{k} c^k \, T(\underbrace{x^r Q, \dots, x^r Q}_{a-k}, \underbrace{t^r Q, \dots, t^r Q}_{k}).$$

A polynomial in c vanishing for all $c \in \mathbb{C}$ has zero coefficients. The coefficient of c^1 is $a\, T(x^r Q, \dots, x^r Q, t^r Q)$ and $a \neq 0$, which gives (8).

Step 2: freeing the last slot. *Claim:* for all $x, t \in V$ and every product $Q = y_1 \cdots y_m$ of m vectors,

$$H_x := T(\underbrace{x^r Q, \dots, x^r Q}_{r \text{ slots}}, t^b) = 0. \tag{9}$$

Fix $x \in V$ and regard $t, y_1, \dots, y_m$ as independent vector variables. For $0 \le k \le m$ put

$$G_k = T(x^r Q, \dots, x^r Q,\ t^{b-k} y_1 \cdots y_k),$$

so $G_0 = H_x$. By Convention 4, G_k is a polynomial, homogeneous of degree $b - k$ in t, of degree $r + 1$ in y_i for $i \le k$, and of degree r in y_i for $i > k$.

(2a) $G_m = 0$. Since $b - m = r$, we have $G_m = T(x^r Q, \dots, x^r Q, t^r Q)$. This is zero for all vector values by (8), hence zero as a polynomial.

(2b) $D_{y_{k+1}, t} G_k = (b - k)\, G_{k+1}$ *for* $0 \le k < m$. The variable t occurs only in the last slot, as $t^{b-k} \cdot (y_1 \cdots y_k)$ with $b - k \ge r + 1 \ge 1$. Lemma 5 applies with $u = t$, $y = y_{k+1}$. In particular

$$D_{y_k, t} \cdots D_{y_1, t} H_x = \frac{b!}{(b-k)!}\, T(x^r Q, \dots, x^r Q, t^{b-k} y_1 \cdots y_k) \qquad (0 \le k \le m). \tag{10}$$

(2c) *Descending induction: $G_{k+1} = 0$ implies $G_k = 0$.* If $G_{k+1} = 0$ then $D_{y_{k+1}, t} G_k = 0$ by (2b). Apply Lemma 7 with $U = t$, $Y = y_{k+1}$ and the other y_i as additional variables. The polynomial G_k has

$$p = \deg_t = b - k, \qquad q = \deg_{y_{k+1}} = r, \qquad p - q = m - k \ge 1.$$

Here y_{k+1} still has degree r, because the earlier operators raised only the degrees of $y_1, \dots, y_k$. Hence $G_k = 0$.

Starting from (2a) and applying (2c) for $k = m - 1, \dots, 0$ gives $G_0 = H_x = 0$ as a polynomial, hence for all vector values.

Step 3: induction on the remaining slots. *Claim:* for all $x, t \in V$,

$$T(x^r Q_1, \dots, x^r Q_r, t^b) = 0 \qquad (Q_1, \dots, Q_r \in \operatorname{Sym}^m V). \tag{11}$$

Fix $x, t \in V$ and define $S_{x,t}(Q_1, \dots, Q_r) := T(x^r Q_1, \dots, x^r Q_r, t^b)$. Multiplication by x^r is a linear map $\operatorname{Sym}^m V \to \operatorname{Sym}^b V$ (as $r + m = b$), so $S_{x,t}$ is a symmetric r-linear form on $\operatorname{Sym}^m V$. By (9), its diagonal vanishes at every product Q of m vectors. By (7), $m \ge 1$ and $m \ge r^2 \ge r(r-1)$, so the induction hypothesis (the proposition for $a - 1 = r$, with m in place of b and the same V) gives $S_{x,t} = 0$.

Step 4: removing the common power. *Claim:* $T(v_1^b, \dots, v_r^b, t^b) = 0$ for all $v_1, \dots, v_r, t \in V$.

Fix $t \in V$ and regard $v_1, \dots, v_r$ and x as independent vector variables. Until now x was a fixed vector; this is legitimate because (11) holds for every $x \in V$. For a tuple $\mathbf{j} = (j_1, \dots, j_r)$ with $0 \le j_i \le r$ put

$$F_{\mathbf{j}} = T\big(x^{j_1} v_1^{b-j_1}, \dots, x^{j_r} v_r^{b-j_r},\ t^b\big).$$

By Convention 4, $F_{\mathbf{j}}$ is homogeneous of degree $b - j_i$ in v_i and of degree $j_1 + \cdots + j_r$ in x. The polynomial of interest is $f := F_{(0,\dots,0)} = T(v_1^b, \dots, v_r^b, t^b)$, of x-degree zero.

(4a) $F_{(r,\dots,r)} = 0$. Since $b - r = m$, this is $T(x^r v_1^m, \dots, x^r v_r^m, t^b)$. It is zero for all vector values by (11) with $Q_i = v_i^m$, hence as a polynomial.

(4b) $D_{x,v_i} F_{\mathbf{j}} = (b - j_i)\, F_{\mathbf{j}+\mathbf{e}_i}$ *whenever* $j_i \le r - 1$, where $\mathbf{e}_i$ is the i-th unit tuple. The variable v_i occurs only in slot i, as $v_i^{b-j_i} \cdot x^{j_i}$ with $b - j_i \ge m + 1 \ge 1$. Lemma 5 applies with $u = v_i$, $y = x$. The derivative does not touch the factors x already present or any other slot.

(4c) *The chain.* Raise j_1 from 0 to r one unit at a time, then j_2, and so on up to j_r. This is a chain of r^2 tuples steps from $(0, \dots, 0)$ to $(r, \dots, r)$, and it corresponds to applying D^r_{x,v_1}, then D^r_{x,v_2}, …, then D^r_{x,v_r}. Multiplying the constants in (4b) gives

$$D^r_{x,v_r} \cdots D^r_{x,v_1} f = \left(\frac{b!}{m!}\right)^r T(x^r v_1^m, \dots, x^r v_r^m, t^b) = 0, \tag{12}$$

because each slot contributes $b(b-1)\cdots(b-r+1) = b!/m!$.

(4d) *Descending induction along the chain:* $F_{\mathbf{j}+\mathbf{e}_i} = 0$ *implies* $F_{\mathbf{j}} = 0$ for each step $\mathbf{j} \to \mathbf{j} + \mathbf{e}_i$ of the chain. If $F_{\mathbf{j}+\mathbf{e}_i} = 0$ then $D_{x,v_i} F_{\mathbf{j}} = 0$ by (4b). Apply Lemma 7 with $U = v_i$, $Y = x$ and the other blocks as additional variables. Before this operator, v_i has been differentiated $j_i \le r - 1$ times and at most $r^2 - 1$ operators have been applied in total, so by (7)

$$p = \deg_{v_i} F_{\mathbf{j}} = b - j_i \ge b - r + 1 = m + 1 \ge r^2 + 1, \qquad q = \deg_x F_{\mathbf{j}} = j_1 + \cdots + j_r \le r^2 - 1.$$

Thus $p > q$ and $F_{\mathbf{j}} = 0$.

Starting from (4a) and descending the chain gives $f = 0$. As t was arbitrary, the claim of Step 4 follows.

Conclusion of the inductive step. By Step 4, $T(w_1, \dots, w_a) = 0$ whenever every w_i is a pure power u_i^b (recall $a = r + 1$). Pure powers span $\mathrm{Sym}^b V$ by Lemma 2, and T is multilinear, so $T = 0$. This completes the induction. □

6 Proof of the theorem and a corollary

Proof of Theorem 1. Let $a \ge 2$ and $b \ge a(a-1)$; in particular $b \ge 1$. By Proposition 8, every symmetric a-linear form on $\mathrm{Sym}^b V$ whose diagonal vanishes on all products of b vectors is zero. By Proposition 6, $\mu_{a,b,V}$ is surjective. □

The ordinary Foulkes conjecture asks only for the existence of an equivariant injection in the opposite direction (see Section 7). Surjectivity gives this in the range of the theorem.

Corollary 9. *For every finite-dimensional complex vector space V, every integer $a \ge 2$, and every integer $b \ge a(a-1)$, there exists a* $\mathrm{GL}(V)$*-equivariant embedding*

$$\mathrm{Sym}^a(\mathrm{Sym}^b V) \hookrightarrow \mathrm{Sym}^b(\mathrm{Sym}^a V).$$

In particular, this gives the sixth symmetric-power comparison for $b \ge 30$.

Proof. The group $\mathrm{GL}(V)$ acts on $\operatorname{Sym} V$ by algebra automorphisms and on $\mathcal{P}$ diagonally, commuting with $\mathfrak{S}_a$. Averaging and multiplication therefore commute with the action, so the surjection $\mu_{a,b,V}$ of Theorem 1 is equivariant. Finite-dimensional polynomial representations of $\mathrm{GL}(V)$ over $\mathbb{C}$ are completely reducible [9, Chapter I, Appendix A, Sections 7–8]. Choose an invariant complement M to $\ker \mu_{a,b,V}$. The restriction $\mu_{a,b,V}|_M$ is an equivariant isomorphism onto $\operatorname{Sym}^a(\operatorname{Sym}^b V)$. Its inverse followed by the inclusion $M \subseteq \operatorname{Sym}^b(\operatorname{Sym}^a V)$ is the required embedding. For $a = 6$ the range is $b \geq 30$. This construction asserts existence only; it does not identify the embedding with a prescribed canonical map. □

Remark 10 (Scope). Lemma 7 holds for any number $n \geq 1$ of coordinates and arbitrary additional variables, and the degree comparisons in Proposition 8 depend only on a and b. Together with the finite-dimensional annihilator argument of Proposition 6, this gives the same bound in every finite dimension, without a stable-dimension reduction. The proof uses positivity of the Fischer Hermitian inner product, the splitting of binary forms over $\mathbb{C}$, and the fact that the scalar factors in the coefficient and differentiation identities (a, $k!$, $b-k$, $b!/m!$) are nonzero in characteristic zero.

7 Context and related work

The statements in this section are background taken from the source, with its citations and locators. They are not used in the proofs above, and they were not checked against the cited works for this rewrite.

Formulation. In positive dimension, Raicu–Sam–Weyman [11, Section 2] formulate the map as $\operatorname{Sym}^b(D^a V) \longrightarrow D^a(\operatorname{Sym}^b V)$, with divided powers realized as invariant tensors. Transporting both sides by averaging gives exactly (2).

The stabilization question. For fixed a and V, one asks for a degree $b_0 \geq 0$ such that $\mu_{a,b,V}$ is surjective for every $b \geq b_0$, either the least such degree in a fixed dimension or a bound valid in every finite dimension. The affine Chow variety of decomposable a-forms consists of products of a linear forms in $\operatorname{Sym}^a(V^*)$. Its coordinate ring identifies with A, whose normalization is R; here b is the coordinate degree [11, Section 2, Theorem 2.3]. See also Landsberg [8, Section 7.6, Propositions 7.11–7.12]. This also explains the reversed roles of the parameters: $T(P, \ldots, P)$ is a homogeneous degree-a polynomial on $\operatorname{Sym}^b V$, tested on products of b vectors viewed as linear forms on V^*; compare the product parametrization in [11, Section 2]. Brion established eventual surjectivity [2] and subsequently gave an effective bound involving both a and the dimension [3, Theorem 3.3]. Eventual surjectivity says that the graded quotient R/A has only finitely many nonzero components; the least stabilization degree is one more than its highest nonzero degree, or zero if $R = A$. Raicu, Sam, and Weyman bound the regularity of the Segre ring and modules of covariants, including the normalization [11, Theorem 3.9]; their theorem does not compute the top degree of R/A. Theorem 1 gives the sufficient bound $a(a-1)$ in every dimension. It answers positively the request for a polynomial bound in [8, Problem 7.19]: in that source's notation $d = b$, $n = a$, $w = \dim V$, the bound is $d \geq n(n-1)$, including when $w = n$. The argument does not determine the least stabilization degree, and the bound is not asserted to be sharp.

Relation to the Foulkes conjecture. The ordinary Foulkes conjecture [5] asks, for $a \leq b$, for a $\mathrm{GL}(V)$-equivariant injection $\operatorname{Sym}^a(\operatorname{Sym}^b V) \hookrightarrow \operatorname{Sym}^b(\operatorname{Sym}^a V)$ [11, Conjecture 2.10]. Corollary 9 gives this in the range of Theorem 1. Existence of an embedding does not specify the canonical

map. At $a = b$ the two representations coincide, so the comparison is automatic, while the canonical endomorphism can fail to be invertible: it has nonzero kernels at $(a, b) = (5, 5)$ and $(6, 6)$ in suitable dimensions [4, arXiv version, Theorem 6(b),(c) and Remark 7], both outside the range of Theorem 1.

Related techniques. Successive weight shifts also occur in Ikenmeyer's proof [7, Section 5] of McKay's propagation theorem [10]; his argument [7, Section 5, Claim 5.1] decomposes tensors according to untouched positions before applying a two-coordinate weight shift. In Ikenmeyer's formulation, the canonical maps in the opposite direction $\mathrm{Sym}^a(\mathrm{Sym}^b V) \longrightarrow \mathrm{Sym}^b(\mathrm{Sym}^a V)$ are considered with $\dim V \geq b$ for each b; for fixed a and $B \geq a$, injectivity at (a, B) propagates to every (a, b) with $b \geq B$ [7, Theorem 1.3 and Section 3]. Here the two derivative chains (Steps 2 and 4) surround an induction on the number of slots and give a surjectivity range directly.

8 Verification checklist

The steps that carry the weight of the proof are listed below, in the order in which a sceptical reader should check them.

1. **Lemma 7, Steps 3–5.** The adjoint of $D_{Y,U}$ for the Fischer inner product is $D_{U,Y}$; the commutator is $D'D - DD' = N_U - N_Y$ (check the sign); and the norm identity (4) follows. Everything else rests on this lemma.

2. **Proposition 8, Step 1.** $(x^r + ct^r)Q$ is a product of b vectors for *every* $c \in \mathbb{C}$ (this uses roots in $\mathbb{C}$), and the coefficient of c is $a\,T(x^rQ, \ldots, x^rQ, t^rQ)$.

3. **Step 2, items (2b)–(2c).** Before the operator $D_{y_{k+1},t}$, the t-degree is $b - k$ and the y_{k+1}-degree is still r (not $r + 1$), so $p - q = m - k \geq 1$. The chain ends at $k = m$, where $b - m = r$ matches (8).

4. **Step 3.** $S_{x,t}$ is a symmetric r-linear form on $\mathrm{Sym}^m V$, and the induction hypothesis is applied with parameters (r, m), which requires $m \geq 1$ and $m \geq r(r - 1)$; both follow from (7).

5. **Step 4, item (4d).** All r^2 operators raise the degree of the same block x, so $q \leq r^2 - 1$, while $p \geq m + 1 \geq r^2 + 1$. This accumulation of degree in the shared block x is what leads to the quadratic sufficient bound; the inequality $p > q$ follows from $m \geq r^2$ (that is, from $b \geq a(a - 1)$). Also check in (4b) that v_i occurs in one slot only, so that Lemma 5 applies.

6. **Proposition 6, Steps B and C.** A_b is spanned by the tensors $P^{\otimes a}$ (this uses Lemma 2 with $k = a$), and $\lambda \mapsto T_\lambda$ is a bijection onto symmetric a-linear forms.

7. **Lemma 2.** The sign count in the polarization identity (3); it is used three times (criterion, base case, final line).

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